

Algebra 1
Unit 7
Lesson 1 Practice

Name Answer key
Date _____
Period _____

For questions 1-2, consider the expression $16y^3 - 8y^2 - 20y$.

1. Write an equivalent expression that uses the greatest common monomial factor.

$$4y(4y^2 - 2y - 5)$$

2. Write an equivalent expression that does not use the greatest common monomial factor. **Answers will vary. Some options:**

$$\begin{array}{ll} 2(8y^3 - 4y^2 - 10y) & 2y(8y^2 - 4y - 10) \\ y(16y^2 - 8y - 20) & \end{array}$$

For questions 3-6, write an equivalent expression for the given expression that uses the greatest common monomial factor.

3. $6f^5g^3 + 12f^2g - 30f^3g^3$

$$6f^2g(f^3g^2 + 2 - 5fg^2)$$

4. $55s^4h^2 + 121s^6h^2 + 11s^2h$

$$11s^2h(5s^2h + 11s^4h + 1)$$

5. $80db - 20d^7b^3$

$$20db(4 - d^6b^2)$$

6. $81x^5y^8 - 18x^2y^3 - 90x^6y^6$

$$9x^2y^3(9x^3y^5 - 2 - 10x^4y^3)$$

For questions 7-10, find the greatest common monomial factor of the expression.

7. $21a^3b^4 - 49ab$

$7ab$

8. $24qc^2 + 20q^5c + 32q^7c^2$

$4qc$

9. $16r^2m^8 - 64r^8m^2 + 40r^3m^3$

$8r^2m^2$

10. $6x^6y^5 - 21x^7y^8 - 36x^2y^4$

$3x^2y^4$